Sandbox for "The output gap does not exist"

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October 13, 2025

1 Weak identification issues

Insight 1 (State space representation with general propagation of structural shocks)**.** *The state space representation with general propagation of structural shocks is given by*

$$\mathbf{s}_t = \mathbf{\Phi}\mathbf{s}_{t-1} + \mathbf{B}\xi_t, \qquad \xi_t \sim \mathcal{N}(0, \mathbf{I}_k) \tag{1}$$

$$\mathbf{y}_t = \mathbf{H}\mathbf{s}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \mathbf{\Sigma}_\varepsilon), \tag{2}$$

where

- $\mathbf{s}_t$ *is the state vector,*
- $\mathbf{y}_t$ *is the observation vector,*
- ξ_t *is the vector of structural shocks which is is normally distributed with mean zero and identity covariance matrix,*
- ε_t *is the vector of measurement noise which is normally distributed with mean zero Gaussian white noise covariance matrix* $\mathbf{\Sigma}_\varepsilon$*,*
- $\mathbf{\Phi}$ *is the transition matrix,*
- $\mathbf{H}$ *is the propagation matrix from states to observables and*
- $\mathbf{B}$ *is the propagation matrix from structural shocks to states.*

Notice that

- In estimation, we work with the reduced form error term in state transition equation (1) given by $\mathbf{v}_t = \mathbf{B}\varepsilon_t$.
- $\mathbf{H}$ and $\mathbf{B}$ are key.

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Insight 2 (Reduced for state space representation)**.** *The reduced form state space representation of Equations (1) and (2) is given by*

$$\mathbf{y}_t = \mathbf{H}\mathbf{\Phi}\mathbf{s}_{t-1} + \mathbf{H}\mathbf{B}\xi_t + \varepsilon_t, \tag{3}$$

where, crucially, $\mathbf{HB}(\mathbf{HB})' = \mathbf{HQH}'$ is the observation space covariance of structural shocks, i.e. how much state innovation variance shows up in observables.

We are now ready to put forth the following claim which will be key to our understanding of what may cause weak identification when working with reduced form state space models to measure output gaps:

Claim 1 (Identification quality of reduced form state space representations)**.** *For reduced form state space representations like Equation (3), the eigenstructure of the Gram matrix $\mathbf{HQH}'$ characterizes the directional observability of structural shocks and hence state identification. A small eigenvalue of $\mathbf{HQH}'$ indicates that some observation direction receives minimal innovation signal, leading to weak Kalman gain and state identification in the corresponding state-space directions.*

General intuition for why eigenvalues of $\mathbf{HQH}'$ matter:

- Shocks are the causes behind business cycles and different gaps.
- $\mathbf{HQH}'$ is the observation space covariance of structural shocks $\implies$ the eigenvalues of this matrix determines the signal of structural shocks in the observation directions.
- The minimum eigenvalue, $\lambda_{\min}(\mathbf{HQH}')$, is the minimum state signal in any observation direction.
- If $\lambda_{\min}(\mathbf{HQH}') \approx 0$ there exists at least one direction in observation space where almost no state signal is seen - only noise. That means that that observation variable which is supposed to help identify the state variables delivers no helpful information and is useless.
- When the weakest observation direction has weak state loading (small $\lambda_{\min}$), the shock signal is negligible and observations are dominated by noise:
  - The Kalman filter "sees" mostly noise in direction
  - It rationally assigns low weight (small Kalman gain) to observations in that direction
  - State estimates don't update much from such observations
  - Filtering uncertainty persists

  This is why $\lambda_{\min}$ predicts identification quality: it measures signal strength relative to noise, and the Kalman filter optimally responds by scaling the Kalman gain accordingly.

- *Note that a Gram matrix, named after the Danish mathematician Jørgen Pedersen Gram, is positive semidefinite and hence all eigenvalues are nonnegative, i.e. $\lambda_{min}(\mathbf{HQH'}) > 0$ $\implies$ when we talk about small eigenvalues this is also true in an absolute sense.*

Sketch proof for Claim 1. The Kalman filter update equations are given by

$$\hat{\mathbf{s}}_{t|t} = \hat{\mathbf{s}}_{t|t-1} + \mathbf{K}_t(\mathbf{y}_t - \mathbf{H}\hat{\mathbf{s}}_{t|t-1})$$
$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1},$$

where $\mathbf{K}_t$ is the Kalman gain. We immediately see that if the Kalman gain is small, estimates barely get updated when new information comes in:

- States barely update: $\hat{\mathbf{s}}_{t|t} \approx \hat{\mathbf{s}}_{t|t-1}$.
- Uncertainty barely decreases: $\mathbf{P}_{t|t} \approx \mathbf{P}_{t|t-1}$.

The Kalman gain is given by

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{S}_t^{-1}, \quad \mathbf{S}_t \equiv \mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}' + \mathbf{\Sigma}_\varepsilon.$$

Let $\mathbf{v}_{\min}$ be the eigenvector of $\mathbf{HQH'}$ which has eigenvalue $\lambda_{\min}(\mathbf{HQH'})$. Note that $\mathbf{v}_{\min}$ is the direction for which the weakest signal is observed.

We have that $\mathbf{HQH'}\mathbf{v}_{\min} = \lambda_{\min}\mathbf{v}_{\min}$ and that $\mathbf{v}'_{\min}\mathbf{HQH'}\mathbf{v}_{\min} = (\mathbf{H'}\mathbf{v}_{\min})\mathbf{Q}(\mathbf{H'}\mathbf{v}_{\min}) = \lambda_{\min}$ which for $\lambda_{\min} \approx 0$ means that $\|\mathbf{H'v}\|$ is small.[a]

Step 1: Decompose $\mathbf{S}_t\mathbf{v}_{\min}$.

$$\mathbf{S}_t\mathbf{v}_{\min} = \mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{v}_{\min} + \mathbf{\Sigma}_\varepsilon\mathbf{v}_{\min}.$$

Notice that the sum contains two terms:

- Signal term: $\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{v}_{\min}$
- Noise term: $\mathbf{\Sigma}_\varepsilon\mathbf{v}_{\min}$.

Step 2: Bound the signal term.
By the Cauchy-Schwartz inequality, we have that

$$\|\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{v}_{\min}\| \leq \|\mathbf{H}\| \cdot \|\mathbf{P}_{t|t-1}\| \cdot \|\mathbf{H}'\mathbf{v}_{\min}\|.$$

Assume that $\mathbf{Q} = \mathbf{I}$, then $\|\mathbf{H}'\mathbf{v}_{\min}\| = \sqrt{\lambda_{\min}}$ is small and we have that[b]

$$\|\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{v}_{\min}\| = \mathcal{O}(\sqrt{\lambda_{\min}}).$$

Step 3. Compare signal to noise.
Without loss of generality, assume $\mathbf{\Sigma}_\varepsilon = \sigma^2\mathbf{I}$. Then

$$\|\mathbf{\Sigma}_\varepsilon \mathbf{v}_{\text{min}}\| = \sigma^2\|\mathbf{v}_{\text{min}}\| = \sigma^2.$$

When λ_{min} is small:

$$\|\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{v}_{\text{min}}\| \ll \sigma^2,$$

and therefore

$$\mathbf{S}_t\mathbf{v}_{\text{min}} = \mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{v}_{\text{min}} + \mathbf{\Sigma}_\varepsilon\mathbf{v}_{\text{min}} \approx \sigma^2\mathbf{v}_{\text{min}},$$

i.e. $\mathbf{S}_t\mathbf{v}_{\text{min}}$ is noise-dominated.

Step 4: Invert.
Since $\mathbf{S}_t\mathbf{v}_{\text{min}} \approx \sigma^2\mathbf{v}_{\text{min}}$ we have that

$$\mathbf{S}_t^{-1}\mathbf{v}_{\text{min}} \approx \frac{1}{\sigma^2}\mathbf{v}_{\text{min}}$$

Step 5: Compute Kalman gain.

$$\mathbf{K}_t\mathbf{v}_{\text{min}} = \mathbf{P}_{t|t-1}\mathbf{H}'\mathbf{S}_t^{-1} \approx \mathbf{P}_{t|t-1}\mathbf{H}'\frac{1}{\sigma^2}\mathbf{v}_{\text{min}}.$$

Step 6: Bound the Kalman gain magnitude.

$$\|\mathbf{K}_t\mathbf{v}_{\text{min}}\| \leq \frac{1}{\sigma^2}\|\mathbf{P}_{t|t-1}\| \cdot \|\mathbf{H}'\mathbf{v}_{\text{min}}\| = \frac{1}{\sigma^2}\|\mathbf{P}_{t|t-1}\| \cdot \sqrt{\lambda_{\text{min}}},$$

and therefore $\|\mathbf{K}_t\mathbf{v}_{\text{min}}\| = \mathcal{O}(\sqrt{\lambda_{\text{min}}})$. □

[a] In the $\mathbf{Q}$-weighted metric sense - should develop this further but it is not an issue that invalidates the points made here

[b] All insights hold when $\mathbf{Q} \neq \mathbf{I}$ but the math gets a little bit more tedious. What we learn from adding $\mathbf{Q} \neq \mathbf{I}$ is that the eigenvalues of $\mathbf{Q}$ also matter and if $\mathbf{Q}$ is ill-conditioned, identification quality also detoriates. $\mathbf{Q}$ may be ill-conditioned if for example structural shocks nearly cancels each others effects on states.

1.1 Context-specific example: weak identification of flex-price gap when Phillips curve slope parameter is low

We have been puzzled by why it is hard to estimate the flex-price gap using simulated data from the basic NK model with cost-push shocks and the following state space specification:

$$\begin{pmatrix} y_t \\ \pi_t - \beta\mathbb{E}_t\pi_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & \kappa \end{pmatrix} \begin{pmatrix} \mu_t^f \\ \psi_t^f \end{pmatrix} + \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^\pi \end{pmatrix},$$

where

- y_t is output,
- π_t is inflation,
- $\mathbb{E}_t\pi_{t+1}$ is inflation expectations,
- μ_t^f is flex-price potential output,
- ψ_t^f is flex-price gap,
- κ is the (reduced form) Phillips curve slope parameter.

We have that

$$\mathbf{H} = \begin{pmatrix} 1 & 1 \\ 0 & \kappa \end{pmatrix},$$

and $\mathbf{Q} = \mathbf{I}_2$.

Let's assume that $\kappa = 0.1$. Then

$$\mathbf{HQH}' = \mathbf{HH}' = \begin{pmatrix} 2 & \kappa \\ \kappa & \kappa^2 \end{pmatrix} = \begin{pmatrix} 2 & 0.1 \\ 0.1 & 0.01 \end{pmatrix}.$$

The eigenvalues are $\lambda_1 = 2.005$ and $\lambda_2 = 0.005 \equiv \lambda_{\min}$. The minimum eigenvalue appears in the direction with eigenvector $\mathbf{v}_{\min} = \begin{pmatrix} 0.0501 \\ -0.9987 \end{pmatrix}$.[1]

What does this mean? $\mathbf{v}_{\min}$ tells us that the weak signal almost exclusively comes from the second observable, $\pi_t - \beta\mathbb{E}_t\pi_{t+1}$. Furthermore, we can compute how weak this signal is compared to the noise. First, we compute the composite observations variable $\tilde{\pi}_t$ by rotating observation space $(y_t, \pi_t - \beta\mathbb{E}_t\pi_{t+1})$ using our eigenvector $\mathbf{v}_{\min}$:

$$\begin{aligned}
\tilde{\pi}_t &= \begin{pmatrix} y_t \\ \pi_t - \beta\mathbb{E}_t\pi_{t+1} \end{pmatrix}' \mathbf{v}_{\min} \\
&= 0.05y_t - 0.9987(\pi_t - \beta\mathbb{E}_t\pi_{t+1}) \\
&= 0.05(\mu_t^f + \psi_t^f + \varepsilon_t^y) - 0.9987(0.1\psi_t^f + \varepsilon_t^\pi) \\
&= \underbrace{0.05\mu_t^f - 0.0499\psi_t^f}_{\text{signal: } \mathcal{O}(\kappa)} + \underbrace{0.05\varepsilon_t^y - 0.9987\varepsilon_t^\pi}_{\text{noise: } \mathcal{O}(.9987)}.
\end{aligned}$$

Hence, we see that the signal coefficients are $\mathcal{O}(\kappa) = \mathcal{O}(0.1)$ while noise coefficients are $\mathcal{O}(0.9987)$. In turn, the signal is around $10\times$ weaker than the measurement noise in the direction dominated by the nominal variables.

[1] Computed using Matlab's `eig` function on $\mathbf{HQH}'$ with $\kappa = 0.1$.